



Course Code & Title	Computational Mathematics II
Programme & School	B.Tech CSE (School of Engineering and Technology)
Faculty Name	Dr. Vishal
Type of Assignment	Handwritten and Upload
Weightage / Marks	10
Submission Mode	LMS

Assignment 1: Differential Equation Modelling

Instructions:

- Solve all questions showing detailed steps.
- Use A4 size sheets.
- Submit neatly written solutions.

Questions:

1. Solve the initial value problem

$$\sin(x) dx + y dy = 0, \quad y(0) = 1.$$

[CO1]

2. Solve

$$(3x^2 + 2xy^2) dx + (2x^2y) dy = 0, \quad y(2) = -3.$$

[CO2]

3. Solve

$$\frac{y}{x^2} + 1 + \frac{y'}{x} = 0.$$

[CO2]

4. Solve

$$(x^2y^2 + y) dx + (2x^3y - x) dy = 0.$$

[CO2]

5. Solve

$$y' - \frac{1}{x}y = x^3, \quad x > 0.$$

[CO2]

6. Solve

$$\frac{dy}{dx} + \frac{y}{x} = y^2.$$

[CO3]

7. Solve

$$\frac{dy}{dx} + \frac{1}{3}y = e^x y^4.$$

[CO3]

8. Solve

$$x \frac{dy}{dx} + y = xy^3.$$

[CO3]

9. Solve

$$\frac{dy}{dx} + \frac{2}{x}y = -x^2 \cos x \cdot y^2.$$

[CO3]

10. Solve

$$3 \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0.$$

[CO4]

11. Solve

$$y'' - 6y' + 13y = 0.$$

[CO4]

12. Solve the initial value problem

$$y'' + y = 0, \quad y(0) = 2, \quad y'(0) = 3.$$

[CO4]

13. Solve the boundary-value problem

$$y'' + 2y' + y = 0, \quad y(0) = 1, \quad y(1) = 3.$$

[CO4]